

Notes: Peters and Taylor (JFE, 2017)

Intangible Capital and the Investment- q Relation

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1 Big picture

This paper asks how the classic neoclassical investment- q relation changes once intangible capital is treated as capital rather than as a current expense. The traditional empirical investment literature usually measures capital as property, plant, and equipment (PP&E), investment as capital expenditures, and Tobin's q as market value divided by physical capital. Peters and Taylor argue that this measurement is increasingly incomplete because modern firms invest heavily in R&D, organization capital, brand capital, human capital, software, customer relationships, distribution systems, and other intangible assets.

The paper's central message is simple:

- Tobin's q explains intangible investment as well as physical investment at the firm level.
- A q measure that includes intangible capital in the denominator, called **total q** , is a better proxy for investment opportunities.
- Intangible investment has a lower q -slope than physical investment, implying higher convex adjustment costs and slower adjustment to investment opportunities.
- The classic q theory works better in firms, industries, and time periods with more intangible capital.
- At the macro level, accounting for intangible capital greatly strengthens the investment- q relation.

Interpretation: The paper does not discard the neoclassical theory of investment. Instead, it says the theory looks weak partly because researchers have been omitting a major form of capital. Once intangible capital is measured and included, the standard q framework becomes more empirically useful.

2 Incremental contribution of the paper

2.1 Relative to the traditional investment- q literature

The traditional literature tests q theory mostly with physical investment and physical capital. This paper extends the test to **total capital**, where

$$K_{it}^{tot} = K_{it}^{phy} + K_{it}^{int}.$$

This changes both the dependent variables and the main explanatory variable. Investment is measured as physical, intangible, and total investment scaled by total capital, and Tobin's q is measured as firm value divided by total capital:

$$q_{it}^{tot} = \frac{V_{it}}{K_{it}^{phy} + K_{it}^{int}}.$$

2.2 Relative to intangible-capital measurement papers

The paper converts expenditures that accounting rules treat as costs into capital stocks. It measures intangible capital as the sum of:

- **knowledge capital**, accumulated from past R&D spending;
- **organization capital**, accumulated from a fraction of SG&A spending;
- externally purchased balance-sheet intangibles.

The incremental measurement contribution is practical: the authors provide a broad, firm-level proxy for intangible capital that can be computed for public firms using Compustat-style data.

2.3 Relative to cash-flow sensitivity papers

The standard investment-cash flow literature treats R&D and SG&A as operating expenses. Peters and Taylor show that this is conceptually inconsistent if those expenditures are investments. Their adjusted cash-flow measure adds back intangible investment net of a tax adjustment. This changes the interpretation of cash-flow sensitivity, especially for physical investment.

2.4 Relative to macro investment studies

The paper shows that the macro investment- q relation improves dramatically when intangible investment and intangible capital are included. The macro evidence is important because one might think intangible capital is only a firm-level accounting issue. The paper shows that it also changes aggregate investment facts.

Interpretation: The paper’s incremental contribution is not only adding another control variable. It changes the core objects in investment regressions: capital, investment, q , and cash flow. The main empirical finding is that these new objects make the old theory work better.

3 Data and measurement

3.1 Firm-level sample

The firm-level data are public US firms from Compustat over 1975–2011. The authors exclude financial firms, utilities, public-service and international-affairs firms, and firms with missing or non-positive key accounting variables. They also require a minimum amount of physical capital and winsorize regression variables to limit the effect of outliers.

3.2 Physical capital

Physical capital is measured as the replacement-cost proxy for PP&E:

$$K_{it}^{phy} = \text{gross property, plant, and equipment.}$$

Physical investment is capital expenditures:

$$I_{it}^{phy} = \text{CAPX}_{it}.$$

3.3 Intangible capital

Intangible capital is

$$K_{it}^{int} = \text{knowledge capital}_{it} + \text{organization capital}_{it} + \text{external intangibles}_{it}.$$

Knowledge capital is accumulated from R&D using a perpetual inventory method:

$$G_{it} = (1 - \delta_{R\&D})G_{i,t-1} + R\&D_{it}.$$

Organization capital is accumulated from a fraction of SG&A:

$$O_{it} = (1 - \delta_{SG\&A})O_{i,t-1} + 0.3 \times SG\&A_{it}.$$

The 0.3 multiplier is the baseline assumption that 30 percent of SG&A represents investment rather than current operating expense.

3.4 Investment rates

The paper's main investment rates are scaled by total capital:

$$\iota_{it}^{phy} = \frac{I_{it}^{phy}}{K_{i,t-1}^{tot}}, \quad \iota_{it}^{int} = \frac{I_{it}^{int}}{K_{i,t-1}^{tot}}, \quad \iota_{it}^{tot} = \iota_{it}^{phy} + \iota_{it}^{int}.$$

Here

$$I_{it}^{int} = R\&D_{it} + 0.3 \times SG\&A_{it}.$$

For comparison, the paper also reports the standard investment measure

$$\iota_{it}^* = \frac{\text{CAPX}_{it}}{K_{i,t-1}^{phy}}.$$

3.5 Tobin's q

The paper compares two main q proxies. The standard proxy is

$$q_{it}^* = \frac{V_{it}}{K_{it}^{phy}},$$

where V_{it} is market value of equity plus book debt minus current assets. The new proxy is

$$q_{it}^{tot} = \frac{V_{it}}{K_{it}^{phy} + K_{it}^{int}}.$$

Interpretation: The usual q^* can be mechanically inflated when firms have large unmeasured intangible capital. Total q lowers this measurement error by including intangible assets in replacement cost.

4 Model and empirical predictions

4.1 Neoclassical investment with two capital goods

The model has a firm choosing physical and intangible investment to maximize value:

$$V_{it} = \max_{I^{phy}, I^{int}} E_t \int_0^\infty \left[\Pi(K_{i,t+s}^{tot}, \epsilon_{i,t+s}) - c_{i,t+s}^{phy} - c_{i,t+s}^{int} \right] e^{-rs} ds.$$

Capital evolves as

$$dK^m = (I^m - \delta K^m)dt, \quad m \in \{phy, int\}.$$

Adjustment costs are quadratic:

$$c_i^m(I^m, K^{tot}, p^m) = p^m I^m + K^{tot} \left[\zeta_i^m \frac{I^m}{K^{tot}} + \frac{\gamma_i^m}{2} \left(\frac{I^m}{K^{tot}} \right)^2 \right].$$

4.2 Key prediction 1: one marginal q

Because profits are homogeneous in total capital, physical and intangible capital share the same marginal q :

$$\frac{\partial V}{\partial K^{phy}} = \frac{\partial V}{\partial K^{int}} = \frac{V}{K^{tot}} = q^{tot}.$$

4.3 Key prediction 2: investment rates are linear in total q

Optimal investment rates satisfy

$$\begin{aligned} \iota_{it}^{phy} &= \frac{1}{\gamma_i^{phy}} \left(q_{it}^{tot} - \zeta_i^{phy} - p_{it}^{phy} \right), \\ \iota_{it}^{int} &= \frac{1}{\gamma_i^{int}} \left(q_{it}^{tot} - \zeta_i^{int} - p_{it}^{int} \right). \end{aligned}$$

Thus both physical and intangible investment should respond to total q , but their slopes can differ because adjustment costs differ.

4.4 Key prediction 3: the q -slope estimates convex adjustment costs

In panel regressions of investment rates on total q with firm and year fixed effects,

$$\iota_{it}^m = \alpha_i + \lambda_t + \beta_m q_{i,t-1}^{tot} + u_{it},$$

the slope is

$$\beta_m = \frac{1}{\gamma^m}.$$

A lower q -slope means higher convex adjustment costs.

4.5 Key prediction 4: omitting intangibles biases the standard regression

If researchers regress $CAPX/PPE$ on q^* , they omit the ratio of physical to total capital from the regression. The paper shows this creates a downward bias in the q -slope, which means the traditional literature overstates adjustment costs for physical capital.

Interpretation: The theory gives a reason for including intangibles in the denominator of q , the denominator of investment rates, and the construction of cash flow. The empirical sections then ask whether the data behave as the theory predicts.

5 Empirical specifications

5.1 OLS investment- q regressions

The baseline OLS specifications regress physical, intangible, and total investment on lagged q , firm fixed effects, and year fixed effects. The main comparison is between regressions using total q and regressions using standard q .

5.2 Bias-corrected regressions

Because q is measured with error, OLS slopes are biased downward. The authors use the Erickson, Jiang, and Whited cumulant estimator to correct measurement-error bias. The estimator also produces two diagnostic statistics:

- ρ^2 : how well the unobservable true q explains investment;
- τ^2 : how well the measured q proxy explains the true q .

5.3 Cash-flow regressions

The paper tests whether cash flow explains investment after controlling for q :

$$\ell_{it}^m = \alpha_i + \lambda_t + \beta q_{i,t-1}^{tot} + \theta c_{it}^{tot} + u_{it}.$$

The theory predicts $\theta = 0$, but empirically, cash flow remains important for some investment measures.

5.4 Subsample and macro tests

The paper repeats the regressions across:

- intangible-intensity quartiles;
- Fama-French industry groups;
- early versus late sample periods;
- aggregate US macro data.

Interpretation: The paper is mostly about measurement and theory-based regression design. It is not a causal identification paper in the narrow treatment-effect sense. Its credibility comes from consistent patterns across OLS, measurement-error-corrected estimates, subsamples, macro data, and robustness checks.

6 Tables and figures

6.1 Table 1. Summary statistics and Figure 1. Capital intangibility over time

Table 1

Summary statistics.
Statistics are based on the sample of Compustat firms from 1975 to 2011. The physical capital stock, K^{ph} , is measured as property, plant, and equipment (PP&E). We estimate the intangible capital stock, K^{in} , by applying the perpetual inventory method to firms' intangible investments, defined as research and development (R&D) and $0.3 \times$ selling, general, and administrative (SG&A) spending. We then add in firms' balance sheet intangibles. Intangible intensity equals $K^{in} / (K^{in} + K^{ph})$. Knowledge capital is the part of intangible capital that comes from R&D. The denominator for all new measures is $K^{in} + K^{ph}$. The denominator for all standard measures is K^{ph} . The numerator for both q variables is the market value of equity plus the book value of debt minus current assets. The numerator for q^{ph} is capital expenditure (CAPX), and the numerator for q^{in} is $R\&D + (0.3 \times SG\&A)$. Total investment $i^{ph} = i^{ph} + i^{in}$. The numerator for standard cash flow is income before extraordinary items plus depreciation expenses. The numerator for total cash flow is the same but adds back intangible investment net a tax adjustment.

Variable	Mean	Median	Standard deviation	Skewness
Intangible capital stock (millions of dollars)	427	41.7	1990	11.6
Physical capital stock (millions of dollars)	1237	77.9	6691	16.5
Intangible intensity	0.43	0.45	0.27	-0.01
Knowledge capital/intangible capital	0.24	0.01	0.37	1.65
New measures				
Total q (q^{ph})	1.11	0.57	1.91	3.76
Physical investment (i^{ph})	0.10	0.06	0.14	3.47
Intangible investment (i^{in})	0.11	0.09	0.11	1.92
Total investment (i^{tot})	0.21	0.16	0.18	2.61
Total cash flow (c^{tot})	0.16	0.15	0.19	0.52
Standard measures				
Standard q (q^s)	3.14	0.93	7.22	4.41
CAPX/PP&E (c^s)	0.19	0.11	0.24	3.52
Standard cash flow (c^s)	0.15	0.16	0.62	-1.63

Table 1. Summary statistics.

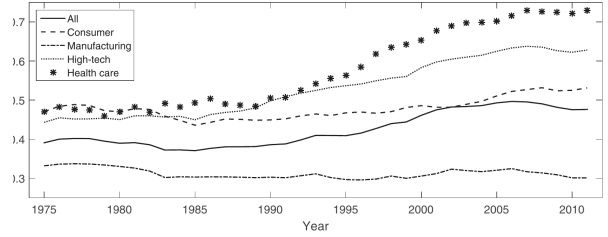


Figure 1. Capital intangibility over time. This figure plots the mean intangible capital intensity over time, both for our full sample and within industry equals $K^{in} / (K^{in} + K^{ph})$, the firm's stock of intangible capital divided by its total stock of capital. We use the Fama and F

Figure 1. Capital intangibility over time.

Read:

- Table 1 shows that the average intangible intensity is 0.43 and the median is 0.45, meaning that almost half of total capital is intangible in the typical firm-year.
- Knowledge capital is only part of intangible capital; organization capital is the dominant component.
- Total q has a much lower mean and dispersion than standard q , because the denominator includes intangible capital.
- Figure 1 shows that intangible intensity rises over time, especially after the early 1990s. High-tech and health care are most intangible-intensive, while manufacturing is lower but still materially intangible.

Economic message: Intangible capital is not a small correction. It is close to half of the capital stock in the firm-level sample. Treating intangibles as zero turns a major economic asset into a measurement error in both q and investment.

6.2 Table 2. Ordinary least squares results and Table 3. Bias-corrected results

Intangible intensity equals $K^{int}/(K^{int} + K^{phy})$, the firm's stock of intangible capital divided by its total stock of capital. We use the Fama and French five-industry definition and exclude industry "Other".

Table 2
Ordinary least squares results.

Results are from OLS panel regressions of investment on lagged Tobin's q and firm and year fixed effects. Each column uses a different investment measure. Physical investment (i^{phy}) equals capital expenditure (CAPEX) scaled by total capital ($K^{tot} = K^{phy} + K^{int}$). Intangible investment (i^{int}) equals research and development (R&D) + 0.3 $\times$ selling, general and administrative (SG&A) expense, scaled by K^{tot} . Total investment equals $i^{phy} + i^{int}$. R&D investment equals R&D scaled by total capital. The R&D column excludes observations with missing R&D. The investment measure in the final column is CAPEX divided by property, plant and equipment (PP&E). Panel A shows regressions on total q , denoted q^{tot} . Panel B shows regressions on standard q , denoted q^s . The numerator for both q variables is the market value of equity plus the book value of debt minus current assets. The denominator for q^{tot} is K^{tot} . The denominator for q^s is K^{phy} . Standard errors clustered by firm are in parentheses. We report the within-firm R^2 . Panel C tests whether the R^2 values in Panels A and B are different, taking into account the correlation across regressions and again clustering by firm. Data are from Compustat from 1975 to 2011.

	Investment scaled by total capital (K^{tot})				
	Physical (i^{phy})	Intangible (i^{int})	Total (i^{tot})	R&D	CAPEX/PPE (i^c)
Panel A: Regressions with total q					
Total q (q^{tot})	0.029 (0.001)	0.020 (0.000)	0.049 (0.001)	0.013 (0.000)	0.062 (0.001)
R^2	0.209 (0.008)	0.279 (0.007)	0.327 (0.006)	0.270 (0.009)	0.244 (0.008)
Panel B: Regressions with standard q					
Standard q (q^s)	0.006 (0.000)	0.005 (0.000)	0.011 (0.000)	0.003 (0.000)	0.017 (0.000)
R^2	0.139 (0.009)	0.266 (0.008)	0.250 (0.010)	0.250 (0.010)	0.233 (0.008)
Panel C: Difference in R^2 (Panel A-Panel B)					
ΔR^2	0.070 (0.003)	0.013 (0.004)	0.077 (0.003)	0.020 (0.005)	0.011 (0.003)
Number of observations	141,800	141,800	141,800	75,426	141,800

Table 2. Ordinary least squares results.

Table 3

Bias-corrected results.

Results are from regressions of investment on lagged Tobin's q , firm fixed effects, and (in Panel B) contemporaneous cash flow, all estimated using the cumulant estimator. Each column uses a different investment measure as defined in Table 2. Total (standard) q equals the firm's market value scaled by K^{tot} (K^{phy}). The numerator for standard cash flow is income before extraordinary items plus depreciation expenses. The numerator for total cash flow is the same but adds back intangible investment net a tax adjustment. Total cash flow is scaled by K^{tot} ; standard cash flow, by K^{phy} . ρ^2 is the within-firm R^2 from a hypothetical regression of investment on true q , and τ^2 is the within-firm R^2 from a hypothetical regression of our q proxy on true q . For comparison, the table also shows the ordinary least squares (OLS) R^2 values from Table 2. Standard errors clustered by firm are in parentheses. Data are from Compustat from 1975 to 2011.

	Investment scaled by total capital (K^{tot})				
	Physical (i^{phy})	Intangible (i^{int})	Total (i^{tot})	R&D	CAPEX/PPE (i^c)
Panel A: Regressions without cash flow					
Total q (q^{tot})	0.070 (0.001)	0.037 (0.001)	0.086 (0.001)	0.023 (0.000)	
Standard q (q^s)					0.036 (0.001)
OLS R^2	0.209 (0.008)	0.279 (0.007)	0.327 (0.006)	0.270 (0.009)	0.233 (0.008)
ρ^2	0.358 (0.008)	0.392 (0.008)	0.423 (0.008)	0.376 (0.011)	0.372 (0.008)
τ^2	0.437 (0.009)	0.559 (0.012)	0.597 (0.010)	0.597 (0.016)	0.492 (0.010)
Panel B: Regressions with cash flow					
Total q (q^{tot})	0.069 (0.001)	0.038 (0.001)	0.086 (0.002)	0.024 (0.001)	
Total cash flow (c^{tot})	0.024 (0.008)	0.050 (0.004)	0.140 (0.009)	0.000 (0.004)	
Standard q (q^s)					0.035 (0.001)
Standard cash flow (c^s)					0.015 (0.004)
OLS R^2	0.235 (0.008)	0.326 (0.007)	0.374 (0.006)	0.281 (0.009)	0.233 (0.008)
ρ^2	0.361 (0.008)	0.447 (0.009)	0.481 (0.008)	0.405 (0.011)	0.371 (0.008)
τ^2	0.435 (0.010)	0.502 (0.014)	0.544 (0.011)	0.568 (0.017)	0.494 (0.011)
Number of observations	141,800	141,800	141,800	75,426	141,800

Table 3. Bias-corrected results.

Read:

- Table 2 shows that total q explains physical investment, intangible investment, and total investment better than standard q does.
- The within-firm R^2 for total investment is higher than for physical or intangible investment alone, consistent with the idea that physical and intangible investment share investment opportunities.
- Table 3 corrects for measurement error in q . The bias-corrected q -slope is lower for intangible investment than for physical investment.
- The lower intangible-investment slope means higher estimated convex adjustment costs for intangible capital.

Economic message: The traditional low q -slope puzzle is partly a measurement problem. Once intangible capital is included, physical capital's estimated adjustment cost falls, while intangible capital appears harder and slower to adjust.

6.3 Table 4. Comparing firms with different amounts of intangible capital and Table 5. Comparing industries

Table 4

Comparing firms with different amounts of intangible capital.

This table shows results from subsamples formed based on yearly quartiles of intangible intensity, which equals the ratio of a firm's intangible to total capital. The first row shows each quartile's mean intangible intensity. Results are from regressions of investment on lagged q , firm fixed effects, and (in Panel E) contemporaneous cash flow. Slopes on q and cash flow, as well as ρ^2 and r^2 values, are from the cumulant estimator. R^2 is from the ordinary least squares (OLS) estimator that includes year fixed effects. Specifications 1-3 use physical (ρ^{ph}), intangible (ρ^{in}), and total investment (ρ^{tot}), respectively, along with total q , all of which are scaled by total capital. Specification 4 uses standard investment (τ^i) = capital expenditure (CAPX) / property plant and equipment (PPE) and standard q (q^i), which is scaled by physical capital. The last row in Panel A shows the ratio of the Specification 2 q -slope to the sum of slopes from Specifications 1 and 2. We conduct inference using the delta method. Specifications 5-8 in Panel E add standard cash flow (c^i) and total cash flow (c^{tot}), defined in Table 3. Standard errors clustered by firm are in parentheses. We use influence functions to conduct inference for ρ^2 and r^2 .

Specification	Quartile 1	Quartile 2	Quartile 3	Quartile 4	Quartile 4 – 1	
					Difference	Standard error
Intangible intensity	8%	33%	56%	76%		
Panel A: Slopes on q						
(1) ρ^{ph} on q^{ph}	0.095	0.081	0.063	0.050	-0.045	(0.004)
(2) ρ^{in} on q^{in}	0.027	0.032	0.035	0.038	0.011	(0.004)
(3) ρ^{tot} on q^{tot}	0.101	0.097	0.086	0.074	-0.027	(0.004)
(4) CAPX/PPE on q^i	0.065	0.052	0.035	0.033	-0.032	(0.007)
$\rho^{ph} / (\rho^{ph} + \rho^{in})$	22%	28%	36%	43%	21%	(2.80%)
Panel B: OLS R^2 values						
(1) ρ^{ph} on q^{ph}	0.219	0.227	0.259	0.284	0.065	(0.026)
(2) ρ^{in} on q^{in}	0.061	0.170	0.306	0.397	0.397	(0.064)
(3) ρ^{tot} on q^{tot}	0.232	0.270	0.357	0.473	0.241	(0.016)
(4) CAPX/PPE on q^i	0.182	0.195	0.248	0.299	0.117	(0.022)
Panel C: ρ^2 values						
(1) ρ^{ph} on q^{ph}	0.261	0.364	0.486	0.612	0.351	(0.027)
(2) ρ^{in} on q^{in}	0.271	0.311	0.377	0.411	0.140	(0.048)
(3) ρ^{tot} on q^{tot}	0.274	0.388	0.498	0.543	0.269	(0.020)
(4) CAPX/PPE on q^i	0.197	0.282	0.379	0.561	0.364	(0.023)
Panel D: r^2 values						
(1) ρ^{ph} on q^{ph}	0.650	0.478	0.374	0.375	-0.275	(0.033)
(2) ρ^{in} on q^{in}	0.196	0.365	0.561	0.792	0.596	(0.031)
(3) ρ^{tot} on q^{tot}	0.664	0.519	0.503	0.659	-0.005	(0.034)
(4) CAPX/PPE on q^i	0.682	0.514	0.483	0.439	-0.243	(0.062)
Panel E: Slopes on cash flow						
(5) ρ^{ph} on q^{ph}, c^{ph}	0.203	0.090	-0.009	-0.036	-0.239	(0.023)
(6) ρ^{in} on q^{in}, c^{in}	-0.018	0.018	0.060	0.110	0.128	(0.013)
(7) ρ^{tot} on q^{tot}, c^{tot}	0.227	0.148	0.100	0.129	-0.098	(0.024)
(8) CAPX/PPE on q^i, c^i	0.182	0.072	0.011	-0.003	-0.185	(0.026)
Number of observations	35,438	35,453	35,442	35,467		

Table 4. Comparing firms with different amounts of intangible capital.

Read:

- Table 4 sorts firms into quartiles of intangible intensity. Total investment R^2 rises sharply from the lowest to the highest intangible quartile.
- The ρ^2 statistic also rises with intangible intensity, meaning the true investment- q relation is stronger in firms with more intangible capital.
- Cash-flow slopes are generally lower in more intangible-intensive subsamples, especially for physical and standard investment measures.
- Table 5 shows similar patterns across industries: high-tech and health care, which are more intangible-intensive, display stronger investment- q fit than manufacturing in several dimensions.

Economic message: The classic q theory works better where intangible capital is more important. This is the opposite of the view that intangible firms are too hard to explain with neoclassical investment theory.

Table 5

Comparing industries.

This table shows results from industry subsamples. We use the Fama and French five-industry definition, excluding the industry "Other." Remaining details are the same as in Table 4.

Specification	Manufacturing	Consumer	High-tech	Health	Quartile 4 – 1	
					Difference	Standard error
Intangible intensity	31%	48%	55%	62%		
Panel A: Slopes on q						
(1) ρ^{ph} on q^{ph}	0.083	0.085	0.059	0.068	-0.015	(0.005)
(2) ρ^{in} on q^{in}	0.038	0.037	0.036	0.040	0.002	(0.003)
(3) ρ^{tot} on q^{tot}	0.097	0.102	0.079	0.084	-0.013	(0.005)
(4) CAPX/PPE on q^i	0.041	0.042	0.033	0.038	-0.003	(0.003)
Panel B: OLS R^2 values						
(1) ρ^{ph} on q^{ph}	0.194	0.239	0.307	0.244	0.050	(0.038)
(2) ρ^{in} on q^{in}	0.206	0.209	0.407	0.281	0.075	(0.031)
(3) ρ^{tot} on q^{tot}	0.258	0.310	0.460	0.362	0.104	(0.024)
(4) CAPX/PPE on q^i	0.186	0.214	0.354	0.258	0.072	(0.031)
Panel C: ρ^2 values						
(1) ρ^{ph} on q^{ph}	0.254	0.397	0.540	0.551	0.297	(0.036)
(2) ρ^{in} on q^{in}	0.321	0.234	0.474	0.376	0.055	(0.030)
(3) ρ^{tot} on q^{tot}	0.294	0.386	0.572	0.521	0.227	(0.028)
(4) CAPX/PPE on q^i	0.206	0.290	0.549	0.545	0.339	(0.029)
Panel D: r^2 values						
(1) ρ^{ph} on q^{ph}	0.557	0.442	0.431	0.319	-0.238	(0.035)
(2) ρ^{in} on q^{in}	0.398	0.485	0.686	0.545	0.147	(0.041)
(3) ρ^{tot} on q^{tot}	0.632	0.539	0.634	0.522	-0.110	(0.036)
(4) CAPX/PPE on q^i	0.655	0.539	0.511	0.365	-0.290	(0.048)
Panel E: Slopes on cash flow						
(5) ρ^{ph} on q^{ph}, c^{ph}	0.171	0.029	-0.033	-0.059	-0.230	(0.032)
(6) ρ^{in} on q^{in}, c^{in}	0.041	0.106	0.059	-0.019	-0.060	(0.017)
(7) ρ^{tot} on q^{tot}, c^{tot}	0.265	0.190	0.090	0.010	-0.255	(0.034)
(8) CAPX/PPE on q^i, c^i	0.083	0.048	0.001	-0.003	-0.086	(0.018)
Number of observations	40,280	36,884	31,680	11,207		

Table 5. Comparing industries.

6.4 Table 6. Comparing time periods and Table 7. Time-series macro regressions

Table 6

Comparing time periods.

This table shows results from the early (1975–1995) and late (1996–2011) subsamples. The 1995 breakpoint produces subsamples of roughly equal size. Remaining details are the same as in Table 4.

Specification	Early	Late	Late – Early	
			Difference	Standard error
Intangible intensity	39%	47%		
Panel A: Slopes on q				
(1) i^{phy} on q^{tot}	0.083	0.062	–0.021	(0.002)
(2) i^{int} on q^{tot}	0.035	0.037	0.002	(0.002)
(3) i^{tot} on q^{tot}	0.100	0.079	–0.021	(0.004)
(4) CAPX/PPE on q^*	0.043	0.033	–0.010	(0.001)
Panel B: OLS R^2 values				
(1) i^{phy} on q^{tot}	0.205	0.208	0.003	(0.018)
(2) i^{int} on q^{tot}	0.190	0.328	0.138	(0.016)
(3) i^{tot} on q^{tot}	0.273	0.357	0.084	(0.013)
(4) CAPX/PPE on q^*	0.209	0.268	0.059	(0.017)
Panel C: ρ^2 values				
(1) i^{phy} on q^{tot}	0.304	0.407	0.103	(0.016)
(2) i^{int} on q^{tot}	0.259	0.497	0.238	(0.018)
(3) i^{tot} on q^{tot}	0.336	0.511	0.175	(0.016)
(4) CAPX/PPE on q^*	0.262	0.479	0.217	(0.016)
Panel D: τ^2 values				
(1) i^{phy} on q^{tot}	0.501	0.423	–0.078	(0.022)
(2) i^{int} on q^{tot}	0.504	0.584	0.080	(0.030)
(3) i^{tot} on q^{tot}	0.595	0.603	0.008	(0.022)
(4) CAPX/PPE on q^*	0.615	0.477	–0.138	(0.026)
Panel E: Slopes on cash flow				
(5) i^{phy} on q^{tot}, c^{tot}	0.109	–0.033	–0.142	(0.017)
(6) i^{int} on q^{tot}, c^{tot}	0.090	–0.033	–0.123	(0.013)
(7) i^{tot} on q^{tot}, c^{tot}	0.256	0.038	–0.218	(0.020)
(8) CAPX/PPE on q^*, c^*	0.074	–0.008	–0.082	(0.009)
Number of observations	69,753	72,047		

Table 6. Comparing time periods.

Table 7

Time series macro regressions.

Results are from 141 quarterly observations from aggregate US data, from 1972Q2 to 2007Q2. Each column uses a different investment measure. Standard q (q^*) equals the lagged aggregate stock and bond market value divided by the physical capital stock. Hall (2001) computes these measures from the flow of funds. Total q includes intangible capital by multiplying physical q by the ratio of physical to total capital. The ratio is from the Corrado and Hulten (2014) aggregate US data. Bond q is constructed by applying the structural model of Philippon (2009) to bond maturity and yield data. These data are from Philippon's website. Newey–West standard errors with autocorrelation up to 12 quarters are in parentheses. Standard errors for the ordinary least squares (OLS) R^2 values are computed via bootstrap.

	CAPX/ PPE (i^*)	Investment scaled by total capital (K^{tot})		
		Physical (i^{phy})	Intangible (i^{int})	Total (i^{tot})
Panel A: Regressions in levels				
Total q (q^{tot})		−0.001 (0.003)	0.019 (0.003)	0.017 (0.003)
Standard q (q^*)	0.002 (0.003)			
OLS R^2	0.035 (0.034)	0.014 (0.030)	0.570 (0.026)	0.610 (0.040)
Panel B: Regressions in levels with bond q				
Bond q	0.061 (0.009)	0.049 (0.011)	0.006 (0.039)	0.055 (0.032)
OLS R^2	0.462 (0.059)	0.347 (0.050)	0.001 (0.013)	0.139 (0.060)
Panel C: Regressions in four-quarter differences				
Total q (q^{tot})		0.007 (0.002)	0.004 (0.001)	0.01 (0.003)
Standard q (q^*)	0.007 (0.002)			
OLS R^2	0.124 (0.057)	0.106 (0.052)	0.096 (0.056)	0.121 (0.060)
Panel D: Regressions in four-quarter differences with bond q				
Bond q	0.056 (0.006)	0.043 (0.005)	0.017 (0.004)	0.060 (0.008)
OLS R^2	0.606 (0.053)	0.620 (0.059)	0.235 (0.074)	0.530 (0.070)

Table 7. Time-series macro regressions.

Read:

- Table 6 compares early and late periods. The later period has higher intangible intensity and generally better q -model fit.
- Table 7 uses aggregate US macro data. Standard q and standard investment perform poorly in levels.
- Once intangible investment and total q are included, macro-level R^2 values become much larger, especially for intangible and total investment.
- Bond q explains some macro variation, especially in differences, but the intangible-capital adjustment still materially changes the message.

Economic message: The rise of intangible capital is not only a cross-sectional firm phenomenon. It changes the aggregate time-series relation between investment and q .

6.5 Figure 2. Investment-q relation in macro data and Table 8. Robustness: what fraction of SG&A expense is investment?

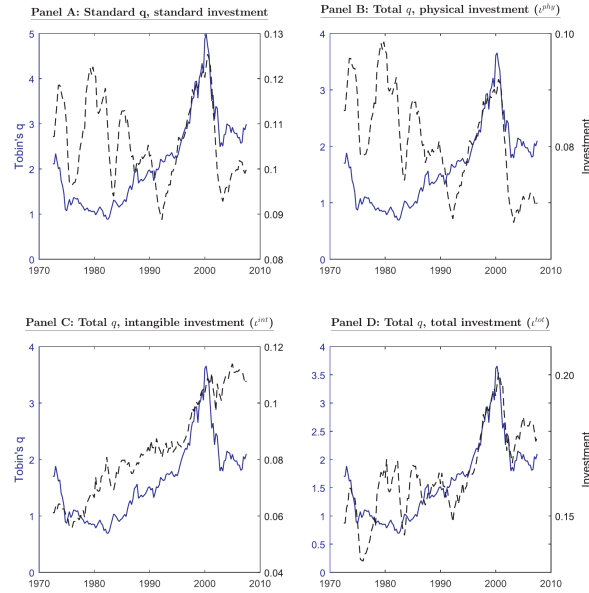


Fig. 2. Investment- q relation in macro data. This figure plots Tobin's q (solid lines) and the investment rates (dashed lines) over time for the aggregate US economy. Panel A uses data from Hali (2001) and shows standard measures that exclude intangible capital. Standard q (q^s) is aggregate market value scaled by the physical capital stock. Standard investment (i^s) equals physical investment scaled by the physical capital stock. Panels B-D also use data from Corrado and Hulten (2014). Total q is aggregate market value scaled by total capital, the sum of the physical and intangible capital stocks. Panel B shows

Figure 2. Investment- q relation in macro data.

Read:

- Figure 2 shows visually why the macro results change: standard investment and standard q do not line up well, but total investment and total q track each other much more closely.
- Table 8 varies the fraction of SG&A treated as investment from zero to one.
- The main conclusions survive a broad range of SG&A multipliers: total q explains total investment best, and intangible investment has a lower q -slope than physical investment.

Economic message: The results are not just an artifact of choosing 30 percent of SG&A as investment. The central patterns remain even when the SG&A multiplier is varied substantially.

Table 8

Robustness: what fraction of SG&A expense is an investment?

Results are from regressions of three investment measures on lagged total q and firm fixed effects. Slopes on total q are from the cumulant estimator. Within-firm R^2 is from the ordinary least squares (OLS) estimator that also includes year fixed effects. The selling, general and administrative (SG&A) multiplier is the fraction of SG&A assumed to represent an investment. Our main analysis uses a 0.3 multiplier. For each multiplier value, we reestimate the intangible investment and capital stocks in the data. Because physical investment, total investment, and total q are scaled by total capital, their values also depend on the SG&A multiplier. Each regression uses 141,800 firm-year Compustat observations from 1975 to 2011. * indicates value used in main analysis

SG&A Multiplier	Investment scaled by total capital (K^{tot})					
	Physical (i^{phy})		Intangible (i^{int})		Total (i^{tot})	
	q -slope	OLS R^2	q -slope	OLS R^2	q -slope	OLS R^2
0.0	0.060	0.223	0.021	0.147	0.064	0.277
0.1	0.064	0.217	0.025	0.256	0.074	0.307
0.2	0.067	0.213	0.032	0.276	0.081	0.320
0.3*	0.070	0.209	0.037	0.279	0.086	0.327
0.4	0.072	0.206	0.043	0.278	0.092	0.331
0.5	0.075	0.203	0.048	0.274	0.097	0.333
0.6	0.077	0.201	0.054	0.270	0.103	0.335
0.7	0.078	0.200	0.06	0.266	0.108	0.335
0.8	0.080	0.198	0.065	0.262	0.113	0.334
0.9	0.082	0.197	0.071	0.257	0.118	0.333
1.0	0.084	0.196	0.075	0.253	0.122	0.332

Table 8. Robustness: what fraction of SG&A expense is investment?

6.6 Table 9. Robustness: alternate measures of intangible capital

Table 9

Robustness: alternate measures of intangible capital.

Results are from regressions of our three main investment measures on lagged total q and firm fixed effects. The column labels indicate the investment measure used. Slopes on q are from the cumulant estimator. We report the within-firm R^2 from the ordinary least squares (OLS) estimator that also includes year fixed effects. The first row reproduces results from Table 3 with our main intangible-capital measure. Rows 2–10 show results using alternate measures of intangible capital. Rows 2 and 3 use alternate values of δ_{SCM} , the depreciation rate for organization capital. Rows 4–6 use alternate values of $\delta_{R\&D}$, the depreciation rate for knowledge capital. Row 7 excludes goodwill from balance sheet intangibles. Row 8 excludes all balance sheet intangibles. Row 9 assumes firms have no intangible capital before entering Compustat, which corresponds to setting $G_0 = 0$ in Eq. (11). Row 10 estimates firms' starting intangible capital using a perpetuity formula that assumes the firm has been alive forever before entering Compustat, as in Falato, Kadyrzhanova, and Sim (FKS 2013). The initial stock of knowledge capital (for example) is $G_0 = R\&D_1 / \delta_{R\&D}$, where $R\&D_1$ is the research and development (R&D) amount in firm i 's first Compustat record. Row 11 uses our main intangible-capital measures but drops each firm's first five years of data. Row 12 uses our main intangible-capital measure but drops firm-year observations with missing R&D. Data are from Compustat from 1975 to 2011.

Specification	Investment scaled by total capital (K^{tot})						N
	Physical (i^{ph})		Intangible (i^{int})		Total (i^{tot})		
	q-slope	OLS R^2	q-slope	OLS R^2	q-slope	OLS R^2	
1. Main results (Table 3)	0.070	0.209	0.037	0.279	0.086	0.327	141,800
2. $\delta_{SCM} = 10\%$	0.071	0.213	0.038	0.294	0.088	0.337	141,800
3. $\delta_{SCM} = 30\%$	0.069	0.207	0.037	0.270	0.086	0.322	141,800
4. $\delta_{R\&D} = 10\%$	0.073	0.209	0.038	0.285	0.089	0.329	141,800
5. $\delta_{R\&D} = 15\%$	0.072	0.209	0.038	0.282	0.088	0.328	141,800
6. $\delta_{R\&D} = 20\%$	0.070	0.208	0.038	0.279	0.087	0.327	141,800
7. Exclude Goodwill	0.070	0.209	0.037	0.282	0.086	0.329	141,800
8. Exclude balance sheet intangibles	0.063	0.199	0.035	0.248	0.079	0.306	141,800
9. Zero initial intangible capital	0.069	0.217	0.040	0.297	0.089	0.343	141,800
10. FKS initial multiplier	0.073	0.193	0.033	0.238	0.084	0.293	141,800
11. Drop first five years per firm	0.069	0.132	0.037	0.153	0.078	0.210	82,174
12. Exclude observations with missing R&D	0.062	0.267	0.036	0.359	0.081	0.411	75,426

Table 9. Robustness: alternate measures of intangible capital.

Read:

- Table 9 varies depreciation rates for knowledge capital and organization capital.
- It excludes goodwill, excludes balance-sheet intangibles, changes the initial stock assumption, and drops early firm-years.
- The qualitative patterns remain: total q explains physical and intangible investment, and total investment is usually best explained by total q .

Economic message: The findings do not depend on one narrow way of constructing intangible capital. The measurement is imperfect, but setting intangible capital to zero is worse.

7 Main results and interpretation

7.1 Result 1: total q is a better proxy for investment opportunities

Total q includes intangible capital in the replacement-cost denominator. This reduces the extreme values and noise that arise when market value is divided only by physical capital. Empirically, total q explains physical, intangible, and total investment better than the traditional q proxy.

Interpretation: If a firm has large intangible assets, market value is high partly because the firm owns intangible capital. Dividing that market value only by physical capital mechanically makes q look too high.

7.2 Result 2: intangible investment is consistent with q theory

The paper finds that intangible investment responds to total q . This is important because the theory was mostly developed and tested for physical investment. The result suggests that intangible capital belongs in the same conceptual framework.

Interpretation: R&D, brand building, and organization investment are not merely discretionary expenses; they behave like investments that respond to opportunities.

7.3 Result 3: intangible capital adjusts more slowly

The lower q -slope for intangible investment implies larger convex adjustment costs. Possible reasons include the difficulty of hiring, training, replacing, or reorganizing workers; the time needed to build organizational routines; and the uncertain nature of knowledge accumulation.

Interpretation: An economy with more intangible capital may adjust more slowly to shocks in investment opportunities.

7.4 Result 4: cash flow remains a challenge

Simple q theory predicts that cash flow should not explain investment after controlling for q . The data do not fully support this prediction. Physical investment remains sensitive to cash flow, and the measurement of SG&A investment makes the cash-flow evidence for intangible investment ambiguous.

Interpretation: Including intangible capital improves the q side of the theory, but it does not eliminate the classic investment-cash flow puzzle.

7.5 Result 5: the theory works better where intangibles matter more

Across intangible-intensity quartiles, industries, and time periods, investment is better explained by q in settings with more intangible capital.

Interpretation: The paper reframes the modern intangible economy as a setting where q theory is more relevant, not less relevant.

8 Identification, limitations, and caveats

8.1 Identification logic

The paper's empirical design is not a randomized or quasi-experimental design. The core identification is theory-based measurement: if the model is correct, physical and intangible investment should both respond to total q , and the slope should identify convex adjustment costs. The evidence is persuasive because the same message appears across multiple specifications, estimators, samples, and robustness checks.

8.2 Key caveats

- The intangible-capital stock is constructed, not directly observed.

- The SG&A investment share is assumed in the baseline and cannot be directly verified firm by firm.
- R&D and SG&A may contain both investment and operating expenses.
- The theory assumes physical and intangible capital can be aggregated into total capital, but in reality they may be complements.
- Public Compustat firms may not represent private firms or the whole economy.
- Cash-flow sensitivity remains difficult to interpret because it may reflect financing constraints, measurement error, or information about marginal q .

Interpretation: The paper’s lesson is best read as a measurement and modeling correction, not as a claim that intangible capital is perfectly observed or that simple q theory is complete.

9 Short summary

Peters and Taylor show that the standard empirical investment- q framework looks different once intangible capital is treated as capital. They construct firm-level intangible capital from R&D, SG&A, and balance-sheet intangibles, define total capital as physical plus intangible capital, and propose total q as firm value divided by total capital. Total q explains physical and intangible investment roughly equally well and total investment even better. Intangible investment has a lower q -slope, implying higher convex adjustment costs. The theory performs better in more intangible-intensive firms, industries, and years. Macro evidence reinforces the same message.

10 Conclusion

The paper’s main contribution is to modernize the empirical investment- q relation for an intangible-capital economy. The standard approach omits a large and growing component of capital, which weakens the measured relation between investment and q and biases adjustment-cost estimates. Once intangible capital is included, the neoclassical theory of investment appears more useful. The paper therefore provides both a practical measurement tool and a conceptual shift: intangible capital should be part of core investment theory, not an accounting footnote.